UNIVERSITY OF WARWICK	
Department	Computer Science
Module Code	CS258
Module Title	Database Systems
Exam Paper Code	CS2580_A
Duration	2 hours
Exam Paper Type	24-hour window

STUDENT INSTRUCTIONS

1. Read all instructions carefully. We recommend you read through the entire paper at least once before writing.

2. There are **FIVE** questions. All candidates should **attempt FOUR questions**.

3. You should not submit answers to more than the required number of questions.

4. You should handwrite your answers either with paper and pen or using an electronic device with a stylus (unless you have special arrangements for exams which allow the use of a computer). Start each question on a new page and clearly mark each page with the page number, your student id and the question number.
Handwritten notes must be scanned or photographed and all individual solutions collated into a single PDF with pages in the correct order.

5. Please ensure that all your handwritten answers are written legibly, preferably in dark blue or black ink. If you use a pencil ensure that it is not too faint to be captured by a scan or photograph.

6. Please check for legibility before uploading. It is your responsibility to ensure your work can be read.

7. Add your student number to all uploaded files.

8. You are permitted to access module materials, notes, resources, references and the internet during the online assessment.

9. You must not communicate with any other candidate during the assessment period or seek assistance from anyone else in completing your answers. The Computer Science Department expects the conduct of all students taking this assessment to conform to the stated requirements. Measures will be in operation to check for possible misconduct. These will include the use of similarity detection tools and the right to require live interviews with selected students following the assessment.

10. By starting this assessment, you are declaring yourself fit to undertake it. You are expected to make a reasonable attempt at the assessment by answering the questions in the paper.

IMPORTANT INFORMATION

- We strongly recommend you use Google Chrome or Mozilla Firefox to access the Alternative Exams Portal.

- You are granted an additional 45 minutes beyond the stated duration of this assessment to allow for downloading/uploading your assessment, your files and any technical delays.

- Students with approved Alternative Exam Arrangements (Reasonable Adjustments) that permit extra time and/or rest breaks will have this time added on to the stated duration.

- You must have completed and uploaded the assessment before the 24-hour assessment window closes.

- Late submissions are not accepted.

- If you are unable to submit your assessment, you must submit Mitigating Circumstances immediately, attaching supporting evidence and your assessment. The Mitigating Circumstances Panel will consider the case and make a recommendation based on the evidence to the Board of Examiners.

SUPPORT DURING THE ASSESSMENT

Operational Support

- Use the Alternative Exams Portal to **seek advice immediately if during the assessment period:**
  - you cannot access the online assessment
  - you believe you have been given access to the wrong online assessment

  Operational support will be available between 09:00 and 17:00 BST for each examination (excluding Sunday)

Technical Support

- If you experience any technical difficulties with the Alternative Exam Portal please contact helpdesk@warwick.ac.uk

Technical support will be available between 09:00 and 17:00 BST for each examination (excluding Sunday)

Academic Support

- If you have an academic query, contact the invigilator (using the 'Contact an Invigilator' tool in AEP) to raise your issue. Please be aware that two-way communication in AEP is not currently possible

  Academic support will normally be provided for the duration of the examination (i.e. for a 2 hour exam starting at 09:00 BST, academic support will normally be provided between 09:00 and 11:45 BST). Academic support beyond this time is at the discretion of the department.

Other Support

- **if you cannot complete your assessment for the following reasons submit Mitigating Circumstances immediately**:

  - you lose your internet connection
  - your device fails
  - you become unwell and are unable to continue
  - you are affected by circumstances beyond your control

CS258

THE UNIVERSITY OF WARWICK

Second Year Examinations: April 2021

Database Systems

Time allowed: 2 hours.

Answer FOUR out of **FIVE** questions.

Continued

1. Relational Algebra and Calculus

   (a) Relational Algebra

   You are given a database whose schema consists of the following table schemas:

   Client (name, age, gender)

   GoesTo (name, restaurant)

   Likes (name, pizza)

   Menu (restaurant, pizza, price)

   Write relational algebra expressions for the following queries:

   i. Retrieve all restaurants to which only females or only males go. [5]

   ii. Retrieve the names of all clients who frequent every restaurant having on its menu at least one pizza they like. [5]

   iii. Find the restaurant serving the cheapest margarita pizza (i.e., pizza = 'margarita'). In the case of ties, return all of the cheapest-pepperoni restaurants. [5]

   (b) Tuple Relational Calculus

   You are given a database whose schema consists of the following table schemas:

   - Student (Sid, Sname, Smajor)
     - Sid is the Primary Key;
     - Smajor is a Foreign Key to Courses.Cdepartment
   - Courses (Cdepartment, Cnumber, Ctitle)
     - (Cdepartment, Cnumber) is the Primary Key;
     - Cdepartment is a Foreign Key to Department.Dshortname
   - Department (Dshortname, Dname, DadminOffice)
     - Dshortname is the Primary Key;
   - Enrollment (Esid, Edept, Enum, Edate)
     - (Esid, Edept, Enum) is the Primary Key;
     - Esid is a Foreign Key to Student.Sid;
     - Enum is a Foreign Key to Courses.Cnumber

   i. Write a tuple relational calculus query to retrieve the student IDs of students of the Business department which are enrolled in "Database Systems" [5]

   ii. Explain with one sentence what the following query does:

   $$\{e.Edate \mid Enrollment(e)\ AND$$
   $$(\exists c)(Courses(c)\ AND\ e.Edept = c.Cdepartment\ AND$$
   $$e.Enum = c.Cnumber\ AND\ c.Ctitle = \text{“}Algorithms\text{”})\}$$ [5]

2. SQL Queries

   (a) Given two tables T1(SSN) and T2(SSN) with only attribute SSN, write an SQL query to retrieve the values in T1 which do not appear in T2.
   Your answer should not use any of the NOT, IN, or EXCEPT operators. [6]

   (b) The TOP operator is used in the SELECT clause of SQL queries in order to restrict the number of tuples in the query answer. For example, the query
   `SELECT TOP (10) T.A FROM T ORDER BY ASC T.A`
   will return the first (top) ten tuples of attribute A (in ASCending order) of table T.
   Now, consider a "Cities" table, with "population" being one of its attributes. Write an SQL query that retrieves the 15th most populous city in the table.
   Your answer should use at most one nested query. [6]

   (c) Consider a table T with two attributes A, B. Write an SQL query that returns those tuples of T which appear more than one time (i.e., the tuple (5, 10) should be returned if and only if T contains the tuple (A=5, B=10) more than one time). [6]

   (d) Consider a table Marks(studentID, mark) where marks are recorded per student. Write a query to retrieve the ID of the student with the 5th highest mark. **Your answer must use self-join**. [7]

Continued

3. Security

(a) A researcher uses a Java program with an SQL database to collect data from a social network. The database contains the following table:

PostContent (authorID, datePosted, messageContent);

The program continuously searches the social network's latest posts, and every time somebody makes a post containing '#sql', the userID of the post's author and the post itself is archived into the researcher's local database. When such a post is found, the system uses the following Java code to insert the content of the post to the PostContent table:

String insQuery = "INSERT INTO PostContent VALUES ("+authorID + ", NOW(), '" + messageContent + "');".

The researcher advertised the study on social media, including details of the database schema. After running this system for a couple of hours, the system is successfully archiving all posts about '#sql'. Then a few hours later, the PostContent table is empty even though there is no evidence of any unauthorized access to the database.

i. Explain what might have gone wrong with the system. Illustrate your answer with a sample of a message that may have been processed by the system. [6]

ii. Explain how the researcher could have protected their data from such a risk, and how this is dealt with by the database drivers and the DBMS. [4]

(b) Another researcher wants to record data about how long the average person spends on their phone in a day. Participants install an app that automatically notifies the research team when the participant's phone is in use. This data is automatically transferred to the researchers and is stored in a secure database with the following schema:

Users (UserID, age, postcode)

PhoneEvents (UserID, ScreenOnTime, ScreenOffTime)

The researcher wants to view summaries of this data with a view that calculates the time that each user spends on their phone. Researchers will then be able to query the data based on postcode; for example finding out how much time people under 18 in 'CV4' spend on their phones.

i. Describe how aggregation can be used to maintain the privacy of individuals, and the circumstances where aggregation might reveal something about an individual and suggest what strategies could be put in place to prevent this. [8]

ii. The researcher also wants to ask a 'Yes'/'No' question to all participants. The answer 'Yes' is considered sensitive.
Describe the technique that could allow the researchers to have a good estimate of the number of people who would answer 'Yes' whilst preserving the privacy of individuals. [7]

4. Normalisation

(a) Consider the relation

w	x	y	z
w1	x1	y1	z1
w2	x2	y1	z2
w2	x2	y2	z2
w3	x3	y3	z3

Which of the following functional dependencies can be said not to hold?

$w \rightarrow x$
$w \rightarrow y$
$\{w,x\} \rightarrow z$
$\{w,x\} \rightarrow \{y,z\}$

Refer to examples in the sample data to illustrate your answers.

[6]

(b) Consider a separate relation with attributes q, r, s, t, u, v. The following functional dependencies are given to hold:

$q \rightarrow s$
$q \rightarrow \{s,t,u\}$
$t \rightarrow v$
$r \rightarrow v$

Fully describe the method for proving that $q \rightarrow v$ [4]

(c) A relation called TrainTimes contains information about a train timetable. The (complex) primary key for TrainTimes is currently (Operator, StartStation, Departing). This represents that a company can only leave a particular station with one train at a particular departure time. CallingAt contains a comma separated list of train stations. CallingTimes is a comma separated list of the times that the train makes those stops, e.g. in the sample below the train operated by MyTrains calls at Rugby at 12:00, Watford at 13:30 and arrives at London Euston at 14:00.

Continued

TrainTimes

Operator	StartStation	ShopsNear Start Station	Departing	CallingAt	CallingTimes
MyTrains	Coventry	3	11:30	Rugby, Watford, London Euston	12:00, 13:30, 14:00
YourTrains	Birmingham	20	18:00	Coventry, Milton Keynes, London Euston	18:20, 19:30, 20:00

i. Describe the design of the TrainTimes relation and any implications when querying this table. [2]

ii. Decompose TrainTimes into two relations such that your design is in the lowest normal form that would ensure that no proper subset of the key determines an attribute that is not a key. Your relations should not contain any new attributes, but you can rename fields if necessary. You can assume that all times can be treated as integers. Illustrate how all the above sample information could be represented in your new design. [6]

iii. Explain the problems that still exist in your new design. [7]

5. SQL JOINs

An organisation operates four museums referred to by the town they are based in: 'Coventry', 'Kenilworth', 'Warwick' and 'Leamington'. Customers buy tickets for the museums, and their details are stored in the schema below.

Customer(CustomerID, CustomerName, CustomerAddress)

CustomerTicket(CustomerID, Museum, TicketType)

TicketPrice(TicketType, Price)

Museum(Museum, OpenAtWeekends)

The 'Museum' field is the name of the Museum, but is simply the name of the town where the museum is located. Some tickets are 'wildcard' tickets, allowing customers to enter any museum, such tickets are denoted with a Museum of NULL.

Prices are stored as integer values representing the number of pence each ticket costs. For example, a ticket costing £5.50 is stored as 550. Some tickets are given away as part of promotional deals and so do not have a price (but do have a TicketType) and are not listed in the TicketPrice table.

The coalesce function can be used to replace nulls with a particular value (or the value of another field). Given a list of input parameters, it will always return the first value that is not null. For instance coalesce(NULL,1) returns 1, and coalesce(a, 5) will return a if a is not null or 5 if a is NULL.

The concat function can be used to concatenate strings together, for instance concat('a', 'b') returns 'ab'.

(a) Write an SQL query that would show a list of all customer names and the museums that they are allowed to visit. If they can visit multiple museums, make sure the result contains a row for each museum they can visit. You must use explicit JOINs at most one subquery (although you may combine queries using set operations). [8]

(b) The museum organisation needs to find the value of all tickets sold. However, it needs to account for those free promotional tickets by assuming they each have a notional value of £5. Write an SQL query that finds the total value of all tickets sold between 1st January 2021 and 31st March 2021 (inclusive). The query should return a single number (in pence, as they are already stored in the TicketPrice table) and should contain at most one subquery. [5]

(c) The database has become corrupted, and some of the rows in the Museum table have been lost. Fortunately, some of this information can be copied from the CustomerTicket table (apart from the OpenAtWeekends attribute, which should be assumed to be NULL until somebody can manually check the database).

Write a single statement that can automatically add any museums that are missing from the Museum table, assuming NULL for OpenAtWeekends if necessary. [6]

Continued

(d) For each customer, list their name and the museum that is in the same town as their address if they have a ticket that is specifically to attend that museum (i.e., not a 'wildcard' ticket) and the ticket costs more than £10. Note that the address field is a full address, not just a town. [6]

End